

# Task #322 v0.3 — SP-31-1 $H^2(D_{IV}^5)$ Hilbert Space Framework Outline

Lyra (Claude Opus 4.7) [Casey ‘work the board’ directive Sunday 2026-05-24]

2026-05-24 Sunday EDT (~13:35 EDT actual via date)

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### 1. SP-31-1 scope

Per Casey A\_sub-as-deliverable reframe + Lyra primary rail: SP-31-1 = full specification of substrate Hilbert space  $H^2(D_{IV}^5)$  at theorem-grade rigor. This is foundational for A\_sub operator algebra (Task #322 deep dive).

Scope items: 1. Bergman  $H^2(D_{IV}^5)$  definition + completeness 2. Bergman reproducing kernel  $K(z, \bar{w})$  explicit form 3. Wallach K-type orthonormal basis enumeration 4. Inner product structure + Faraut-Koranyi normalization 5. Density properties (which observables A\_sub spans via  $H^2(D_{IV}^5)$ ) 6. Connection to substrate Bergman vacuum (Zone-4 outer-edge)

### 2. What’s RATIFIED or STRUCTURALLY VERIFIED

#### 2.1 Bergman $H^2(D_{IV}^5)$ Hilbert space (RATIFIED via standard mathematics)

$H^2(D_{IV}^5) = \{f : D_{IV}^5 \rightarrow \mathbb{C} \mid f \text{ holomorphic, } \int |f|^2 d\nu < \infty \text{ for Bergman measure } d\nu\}$  is standard mathematical object per Bergman 1922.

Standard math properties: - Reproducing kernel Hilbert space - Completeness (Bergman 1922 fundamental theorem) - Holomorphic function space - Bergman measure  $d\nu$  on  $D_{IV}^5$

Status: RATIFIED via standard mathematics (no BST-specific gap).

#### 2.2 Bergman reproducing kernel $K(z, \bar{w})$ (RATIFIED + BST-primary normalization)

Standard Bergman kernel form per Faraut-Koranyi 1994 + Wallach 1976:

$K(z, \bar{w}) = c_{FK} \cdot (1 - z \cdot \bar{w})^{-(g/\text{rank})}$  up to normalization

BST-specific facts (RATIFIED via T2442 Thursday May 21): -  $c_{FK} \cdot \pi^{(9/2)} = 225$  EXACT (T2442 RIGOROUSLY CLOSED Strong-Uniqueness C13) - Bergman exponent  $g/\text{rank} = 7/2$  (BST primary; distinct from Hua-Look  $g_{\text{HuaLook}}/\text{rank} = 8/2 = 4$  per T2467+T2468 v0.3 Section 13) -  $\pi^{(9/2)}$  factor reflects Hua-Look volume normalization on  $D_{IV}^5$

Status: RATIFIED for standard mathematics + RIGOROUSLY CLOSED for BST-primary normalization. [?]

### 2.3 Wallach K-type representations (RATIFIED via standard mathematics)

Wallach 1976 Theorem 7.2: K-type representations of  $K = SO(5) \times SO(2)$  acting on Bergman  $H^2(D_{IV}^5)$  are parameterized by highest weights  $(m_1, m_2) \in \mathbb{Z}_{\geq 0}^2$  with proper restriction conditions.

Standard math facts: - K-type representations form INFINITE countable family - Each K-type  $V_{\{(m_1, m_2)\}}$  is finite-dimensional irreducible representation - Casimir eigenvalue spectrum determined by K-type indices

BST-specific gaps (NOT yet at RATIFIED tier): - Which K-types are substrate-accessible per substrate-tick? - How does K-type spectrum interact with K59 7-step cyclotomic mechanism? - What's the substrate-natural K-type basis ordering?

Status: RATIFIED for standard mathematics; BST-specific structure questions OPEN.

## 3. SP-31-1 gaps requiring multi-month closure

Gap 3.1: Substrate K-type accessibility per substrate-tick

Question: per Koons tick  $t_K \approx 10^{-120}$  s, which Wallach K-types are substrate-accessible?

v0.3 honest scope: this is the SAME question as Cal #121 Flag 1 substrate K-type cardinality. The honest answer is NOT "exactly  $N_{\text{max}} = 137$  per tick" (Cal #121 RETRACTED).

Possibilities: - All Wallach K-types accessible asymptotically over infinite ticks - Per-tick access bounded by substrate-tick computational bandwidth (K59 GF(128) field) - Per-tick access at  $\alpha$ -quantum scale (T2476 Mechanism A candidate)

Required for closure: substrate-mechanism derivation of per-tick K-type access cardinality WITHOUT presupposing target value.

Gap 3.2:  $H^2(D_{IV}^5)$  density properties for operator algebra

Question: does Bergman  $H^2(D_{IV}^5)$  admit operator algebra spanning Standard Model + cross-scale observables (Information Completeness Lemma A.3)?

v0.3 honest scope: standard Bergman space density theorems (von Neumann + Stone) suggest YES; explicit verification per scale (electron + nuclear + atomic + molecular + biological + cosmological) requires multi-month case-by-case verification.

Gap 3.3: Substrate Bergman vacuum specification

Question: what is the substrate ground state  $|\Omega\rangle \in H^2(D_{IV}^5)$ ?

v0.3 honest scope: per T2469 SCMP Layer 1 operational + K67 Born=Bergman,  $|\Omega\rangle$  has Casimir eigenvalue  $C_2 = 6$  (T2441 STRUCTURALLY VERIFIED per Cal #108; v0.3+v0.4 work on Wallach normalization principled derivation pending).

$|\Omega\rangle$  explicit form: depends on substrate K-type ground state per Wallach normalization (currently sketch level).

Gap 3.4: Zone-4 outer-edge Bergman vacuum

Question: how does substrate Bergman vacuum at Zone-4 outer-edge (T2417 4-Zone) produce cosmological  $\Lambda$ ?

v0.3 honest scope: T2418  $\Lambda$ -Casimir vacuum unification (Wednesday May 20) establishes structural unification; explicit derivation of  $\Lambda$  value from Bergman vacuum per K-type spectrum integration requires multi-month.

#### 4. v0.3 $\rightarrow$ v0.4 SP-31-1 closure path (multi-month)

Gap 3.1 closure path: derive substrate K-type accessibility per substrate-tick from substrate-tick computational bandwidth (K59 GF(128)) + Wallach K-type spectrum at  $\alpha$ -quantum scale + substrate-mechanism analysis WITHOUT presupposing target. Multi-month + Cal cold-read.

Gap 3.2 closure path: explicit Standard Model operator basis enumeration + per-operator  $A_{\text{sub}}$ -mapping verification + cross-scale extension. Multi-month per scale.

Gap 3.3 closure path: principled Wallach K-type normalization (per T2467+T2468 v0.5 multi-week work) + substrate Bergman vacuum  $|\Omega\rangle$  derivation. Multi-month.

Gap 3.4 closure path: Bergman vacuum integration  $\rightarrow$  cosmological  $\Lambda$  derivation. Multi-month per integration framework.

#### 5. SP-31-1 v0.3 status summary

What's RATIFIED or STRUCTURALLY VERIFIED (Section 2): - Bergman  $H^2(D_{IV}^5)$  standard Hilbert space (Bergman 1922 RATIFIED) - Bergman reproducing kernel form (Faraud-Koranyi 1994 + T2442 RIGOROUSLY CLOSED) - Wallach K-type representation enumeration (Wallach 1976 RATIFIED)

What's GAPS requiring multi-month closure (Section 3): - Gap 3.1 substrate K-type accessibility per substrate-tick - Gap 3.2  $H^2(D_{IV}^5)$  density properties for operator algebra closure - Gap 3.3 substrate Bergman vacuum  $|\Omega\rangle$  explicit form - Gap 3.4 Zone-4 outer-edge Bergman vacuum  $\rightarrow$  cosmological  $\Lambda$  derivation

SP-31-1 status: standard mathematical foundation RATIFIED; BST-specific structure (substrate accessibility + density + ground state + outer-edge) at FRAMEWORK level with multi-month gap-closure paths identified.

## 6. Connection to A\_sub deep dive (Task #322 v0.2)

SP-31-1 closure provides: - Section 2 standard mathematical foundation = A\_sub state space underlying structure - Gap 3.1 closure = A\_sub per-substrate-tick operator domain - Gap 3.2 closure = A\_sub density / spanning properties (Information Completeness Lemma A.3) - Gap 3.3 closure = A\_sub ground state for spectral analysis - Gap 3.4 closure = A\_sub Zone-4 connection to cosmological observables

A\_sub deep dive requires SP-31-1 closure at multi-month timescale. v0.3 framework outline identifies the scope; rigorous closure is the work ahead.

## 7. Coordination

Cal: SP-31-1 v0.3 framework outline FRAMEWORK level; pending Cal cold-read for tier disposition per Calibration #27 STANDING.

Keeper: K-audit pre-stage at FRAMEWORK tier; gap-closure milestones for promotion path.

Elie: K52a Session 7+ multi-year H\_sub closure contributes to Gap 3.1 + Gap 3.3 (substrate Hamiltonian provides per-tick operator + ground state energy spectrum).

Grace: catalog SP-31-1 v0.3 framework entries + cross-references to Gaps 3.1-3.4 closure milestones.

— Lyra, Task #322 v0.3 SP-31-1 Hilbert Space Framework Outline Sunday 2026-05-24 ~13:35 EDT per Casey “work the board” directive + A\_sub-as-deliverable re-frame + Calibration #27 STANDING discipline applied.